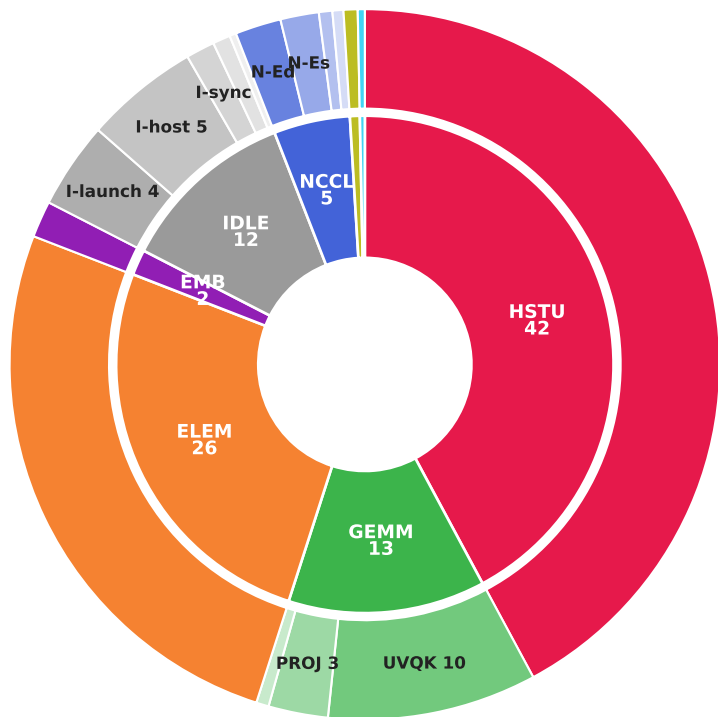


§8b head\_dim 256·4 vs 128·8, same shape (N·D=1024) — GPU-time sunburst, 1B/4096/lognormal bs32 (GB300 8-GPU, 80-step aggregate)

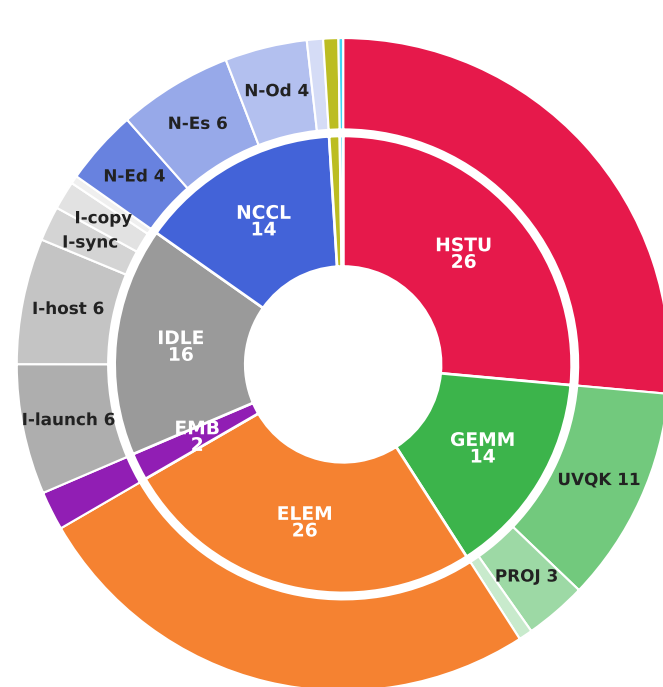
inner ring = grouping · outer ring = leaves · % of step; DISK AREA  $\propto$  step time (d128·8 = 0.84× area) → equal area = equal absolute ms

step −15.9% · the gap is the red HSTU wedge (57 → 30 ms) · both configs move the same bytes — the faster d128·8 config just hides less of its NCCL under compute

CUTLASS d256 · heads=4 (PR #18)  
median 136.1 ms/step · MFU 19.2%



d128 · heads=8 (same shape N·D=1024)  
median 114.5 ms/step · MFU 25.7%



|                            |                       |                                           |                                                 |                            |                    |
|----------------------------|-----------------------|-------------------------------------------|-------------------------------------------------|----------------------------|--------------------|
| ■ HSTU = hstu fwd/bwd      | ■ G-O = gemm / others | ■ N-Ed = nccl DENSE/all-reduce (exposed)  | ■ N-Os = nccl sparse (overlap)                  | ■ I-sync = idle: sync wait | ■ OTH = others     |
| ■ UVQK = gemm / uvqk       | ■ ELEM = elementwise  | ■ N-Es = nccl SPARSE/all-to-all (exposed) | ■ I-launch = idle: kernel-launch (CPU dispatch) | ■ I-copy = idle: H↔D copy  | ■ OVL = overlapped |
| ■ PROJ = gemm / projection | ■ EMB = embedding op  | ■ N-Od = nccl dense (overlap)             | ■ I-host = idle: host/python gap                |                            |                    |